

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CVC 221 — Past Questions & Materials
by Renance

Course: CVC 221
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 200

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Paper 1 of 1 — CVC 221 (Basic Civic Engineering)

Level 200 • Second Semester • Handout

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF CIVIL ENGINEERING
SECOND SEMESTER EXAMINATION 2022/2023 SESSION

COURSE CODE: CVE221 (2 CREDIT UNITS)
COURSE TITLE: BASIC CIVIL ENGINEERING **TIME ALLOWED: 2hrs.15mins**

INSTRUCTION: Answer four (4) questions only at least one question from each section.

Section A: Engineering Survey

1. Engineering survey is an important branch of engineering. Discuss. 17.5 marks

2. The following readings were obtained during an engineering survey project:
 2.425, 1.530, 0.815, 3.450, 2.635, 2.170, 1.635, 0.825, 0.335, 1.630, 2.125, and 3.430. The instrument was shifted after the fourth and eighth readings. The last reading was taken on the Benchmark having a reduced level of 105.2 m and the reading interval 20 m. Using rise and fall method, find the reduced level of all the points. 17.5 marks

Section B: Soil Mechanics

3/ a) Briefly explain the following:
 i) Residual soil ii) Transported soil iii) Mineralogy iv) Porosity
 v) Water content 10 marks

b) A soil sample has a bulk density, ρ_b of 1.76 Mg/m³ and moisture content 8.5%. Calculate:
 i) Void ratio ii) Degree of saturation iii) Porosity 7.5 marks

[take $G_s = 2.7$; $\rho_w = 1000 \text{ kg/m}^3$]

4/ The following are the results of a sieve analysis

Sieve no.	Mass of soil retained (g)
4	0
8	30
10	50
20	79
30	130
40	111
50	200
80	46
Pan	2

Section	B.S	I.S	F.S	Rise	Fall	RL	Remarks
						105.2	
	2.425	1.530		0.895			
		0.815		0.715			
		3.450	3.450	0.61	1.025		
	2.635	2.170		0.465			
		1.635		0.825	1.000	105.2	
	0.335	1.630		1.295			
		3.430	3.430	1.81	2.104	105.2	
	5.395	7.78	7.705				

a) Compute your table and plot the particle-size distribution curve. 12 marks

b) Calculate the coefficient of uniformity, C_u of the sample. 5.5 marks

Section C: Engineering Geology

5/ Discuss the applications of engineering geology to engineering project. 17.5 marks

6. a) What is Stratigraphy? 2.5 marks

b) Mention and explain with diagram three (3) types of faults in rocks 15 marks